
SFTQWEN: SUPERVISED FINE-TUNING OF A SMALL LANGUAGE MODEL FOR DOCUMENT SUMMARIZATION

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ABSTRACT

This work presents **SFTQwen**, an end-to-end pipeline for fine-tuning a Small Language Model (SLM) for document summarization under strict computational constraints. Starting from an unannotated corpus of news articles, high-quality synthetic summaries are generated using a zero-shot Qwen2.5-7B-Instruct teacher model with 4-bit quantization, ensuring compatibility with a Google Colab free-tier GPU. A carefully filtered subset of documents and summaries is then used to supervise the fine-tuning of a compact Qwen2.5-1.5B model via parameter-efficient QLoRA training. The pipeline emphasizes reproducibility, memory efficiency, and robustness to noisy supervision. Experimental results demonstrate that meaningful summarization performance can be achieved by a small model trained entirely on automatically generated labels.

Keywords Supervised Fine-Tuning · Summarization · Small Language Models · LoRA · Quantization

1 Introduction

Document summarization is a core capability for modern language models, yet most state-of-the-art approaches rely on large-scale models and extensive computational resources. In many practical settings, such constraints are prohibitive, motivating the use of Small Language Models (SLMs) adapted to specific tasks.

This project explores whether a compact generative model can be effectively fine-tuned for summarization using only synthetic supervision and limited hardware, following a realistic and fully reproducible training pipeline.

2 Related Work

Abstractive document summarization has traditionally relied on *encoder-decoder* architectures trained with human-annotated datasets, such as Pointer-Generator Networks [1]. While effective, these models typically require substantial computational resources and large-scale supervision.

More recently, decoder-only language models have shown strong zero-shot summarization capabilities, but efficient task adaptation under hardware constraints remains challenging. QLoRA [3] addresses this issue by enabling low-rank adaptation of quantized models, drastically reducing GPU memory requirements while preserving performance.

In parallel, synthetic supervision has emerged as a scalable alternative to manual annotation. Approaches such as Self-Instruct [2] demonstrate that teacher-student pipelines can align models to downstream tasks using automatically generated labels. This work builds on these ideas by combining synthetic summaries with QLoRA-based fine-tuning of a compact decoder-only model.

3 Synthetic High-Quality Label Generation

3.1 Document Selection and Preprocessing

The starting point of the pipeline is an unannotated news corpus derived from the CNN/DailyMail dataset. To reduce task ambiguity and stabilize generation quality, documents were filtered based on their length, retaining only articles between 800 and 1700 characters. This range ensures sufficient informational content while avoiding excessively long contexts that could degrade generation quality.

Additional preprocessing steps were applied to remove formatting artifacts, boilerplate text, and footer content commonly found in news articles, resulting in cleaner and more homogeneous inputs.

3.2 Zero-Shot Teacher Annotation

Summaries were generated using the pretrained Qwen2.5-7B-Instruct model in a zero-shot setting. To comply with hardware constraints, the model was loaded using 4-bit NF4 quantization, allowing inference on a single Colab GPU without loss of stability. A constrained instruction prompt was used to enforce concise summaries limited to a maximum of three sentences.

To improve label quality, generated summaries were filtered based on word count, retaining only outputs containing between 30 and 100 words. This filtering step removes both under-informative summaries and overly verbose generations, which are common failure modes in zero-shot summarization. The resulting dataset consists of approximately 5,000 document-summary pairs, forming a high-quality pseudo-supervised training corpus.

4 Supervised Fine-Tuning

4.1 Model and Training Setup

The student model is Qwen2.5-1.5B, a compact decoder-only language model well-suited for instruction-following tasks. Fine-tuning is performed using QLoRA, where the base model is frozen and loaded in 4-bit precision, while low-rank adapters are injected into attention and feed-forward layers. A LoRA rank of $r = 64$ with dropout is used to balance expressive capacity and regularization, mitigating overfitting to noisy synthetic labels.

Thanks to the Unsloth optimization framework, this QLoRA setup remains highly memory-efficient despite the large LoRA rank and a per-device batch size of 8. The peak reserved GPU memory reaches only 6.15 GB (39.8% of a 16 GB GPU) while updating approximately 4% of the model parameters (about 66 million trainable parameters). Unsloth achieves this by optimizing attention kernels, gradient checkpointing, and memory layouts, substantially reducing activation and optimizer overhead. Without Unsloth, an equivalent configuration would typically require 11–12 GB of VRAM under the same conditions, significantly limiting batch size or context length.

Given the relatively low GPU memory footprint observed during training, it would have been feasible to increase the numerical precision of the base model (e.g., 8-bit quantization using INT8) without exceeding hardware constraints. However, 4-bit quantization was deliberately retained to maximize training and inference speed and to preserve a lightweight deployment profile.

4.2 Training

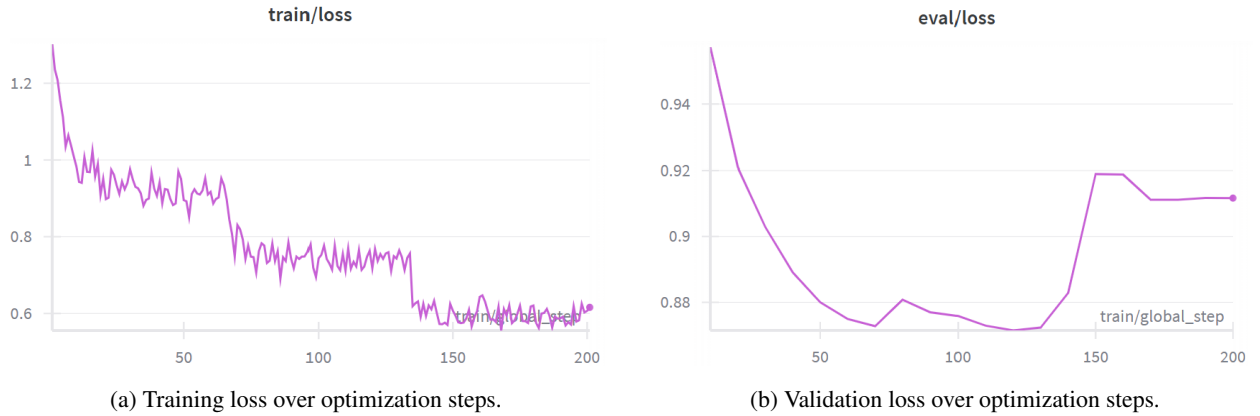


Figure 1: Evolution of training and validation loss during supervised fine-tuning.

Training is performed for three epochs using gradient accumulation and an 8-bit AdamW optimizer. As shown in Figure 1, the training loss decreases steadily, indicating stable convergence under the QLoRA setup despite aggressive quantization. The validation loss follows a similar downward trend during the early and mid stages of training, suggesting effective generalization from synthetic supervision. After approximately 140 optimization steps, a divergence between training and validation loss appears, clearly indicating overfitting, which is expected given the limited dataset size and noise introduced by automatic annotation.

5 Results

The fine-tuned model is evaluated using standard automatic summarization metrics on the validation set. Results are reported in Table 1.

Table 1: Validation performance of SFTQwen.

Metric	Score
ROUGE-1	0.361
ROUGE-2	0.120
ROUGE-L	0.242
METEOR	0.291
BERTScore (F1)	0.307

While these scores remain below those reported by large-scale models trained on human-annotated datasets, they are consistent with prior work on small models trained under constrained settings. In particular, the results demonstrate that a compact model can learn meaningful summarization behavior from purely synthetic supervision, despite limited capacity and noisy labels.

6 Use of AI Assistants

AI assistants, including ChatGPT, were used as supportive tools throughout this project. Their primary role was to assist with code structuring, debugging, and reasoning about best practices for fine-tuning under hardware constraints. Importantly, all methodological choices such as dataset splitting, model selection, and evaluation strategy were made deliberately and critically assessed by the author.

Rather than replacing the problem-solving process, AI tools served as accelerators for iteration and clarification, but I actively sought a balance that maximizes personal learning. Over-delegating to generative AI can significantly reduce

the depth of understanding gained from such projects, whereas learning by building is one of the main objectives of these challenges. In this workflow, AI assistance complemented but never replaced hands-on experimentation, critical thinking, or direct engagement with the underlying tools. Instead, it was used to offload repetitive tasks, enabling greater focus on higher-level reasoning and deeper understanding of the problem.

7 Conclusion

This work shows that meaningful document summarization can be achieved by fine-tuning a Small Language Model under strict computational constraints. By combining high-quality synthetic labels generated by a zero-shot teacher model with parameter-efficient QLoRA fine-tuning, **SFTQwen** attains consistent summarization performance while remaining lightweight and reproducible on a single free-tier GPU. The results highlight the viability of synthetic data generation pipelines as a practical alternative to large-scale supervised training when human annotations are unavailable. Nonetheless, performance remains bounded by label noise and limited model capacity, suggesting that future work should focus on improved synthetic labeling strategies, selective human validation, and preference-based or distillation-oriented fine-tuning.

References

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